

KernelPack RAG — Retrieval Eval Baseline

Experiments v1–v3

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Overview

Four retrieval versions were evaluated against a 10-query eval set covering `kernelpack.solvers` (4 API, 3 workflow, 3 conceptual queries). Each version isolates one change. The pipeline is fixed across all versions: tree-sitter AST chunking at function boundaries, UniXcoder dense embeddings (ChromaDB, L2), BM25 sparse leg, RRF fusion ($k = 60$), top-3 returned.

Note: QA pairs are Codex-generated and have not yet been validated by a domain expert. Recall numbers are directional only.

Results

Version	Change	Hypothesis	recall@3
v1	Baseline — raw source, <code>MIN_LINES=5</code>	—	1/10 (10%)
v2	<code>MIN_LINES=1</code>	recover short-function targets	2/10 (20%)
v3a	Solvers-only scope, no summaries (control)	smaller index = less noise	2/10 (20%)
v3	Solvers-only + LLM summaries prepended	summaries close vocabulary gap	3/10 (30%)

Table 1: Recall@3 progression across retrieval versions.

Wide Retrieval Diagnostic (top-50 dense-only)

Ranks reported from dense-only wide retrieval across v1, v2, and v3. A dash (—) means the target was not found in the top-50. v3a is omitted — it was a recall@3 control only and no wide diagnostic was run. All versions except v2 use `MIN_LINES=5`; targets shorter than 5 lines are excluded from those indexes entirely.

Target	v1	v2	v3	Category
PoissonSolver.last_solve_used_nullspace	—	—	—	Excluded (MIN_LINES)
VariablePoissonSolver.solve	40	—	18	Reranker problem
NonlinearVariablePoissonSolver.get_last_residual_norm	—	2	—	Excluded (MIN_LINES)
PUSLAdvectionSolver.get_solve_stats	—	—	—	Not indexed
DiffusionSolver.set_state_history	1	7	4	Hit
MultiSpeciesDiffusionSolver.set_initial_state	4	7	2	Hit
PUSLFDAdvectionDiffusionSolver.bdf3_step	2	7	6	Hit
build_stencil_properties	2	2	1	Hit
pu_patch_weight	29	42	1	Hit
IncompressibleEulerBDFBackend._get_cached_system	15	19	7	Hit

Table 2: Wide retrieval diagnostic for v1, v2, and v3. Dashes indicate target not found in top-50 dense retrieval.

Key Findings

1. **Vocabulary gap is not the primary bottleneck.** After LLM summary enrichment (v3), 8/10 targets appear in the top-10 retrieved neighborhood. The bottleneck is ranking, not retrieval coverage.
2. **MIN_LINES is the wrong filter.** Lowering the threshold from 5 to 1 (v2) recovered one target (`NonlinearVariablePoissonSolver.get_last_residual_norm`, rank 2) but degraded ranks for five previously-stable targets by introducing short-function noise into the embedding space. This target is a one-liner and is excluded from all versions that use `MIN_LINES=5`.
3. **Scoping alone does not improve recall.** v3a (solvers-only, no summaries) matched v2 at 2/10 — the smaller index reduced noise but did not move targets into the top-3 window on its own.
4. **LLM summaries improve math-heavy targets most.** `pu_patch_weight` moved from rank 42 (v2) to rank 1 (v3). Summary enrichment gives the embeddings plain-English signal to match against natural language queries.

Next Steps

Cross-encoder reranker (v4). With 8/10 targets in the top-10 window, a cross-encoder re-scoring each candidate pair against the query would plausibly recover the buried targets. A chunking strategy that does not rely on line count as a proxy for chunk quality is also warranted — `MIN_LINES` excludes real targets and is not a reliable filter.